

HAORAN ZHANG

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EDUCATION

University of Texas at Austin (UT-Austin) August 2025-May 2029 (expected)
Ph.D. in Electrical and Computer Engineering, Supervisor: Prof. Haris Vikalo

Carnegie Mellon University (CMU) May 2025
M.S. in Electrical and Computer Engineering (Advanced Study), Supervisor: Prof. Carlee Joe-Wong

Huazhong University of Science and Technology (HUST) June 2023
B.E. in Automation (Advanced Class)

The Technical University of Munich (TUM) April 2023 - August 2023
Exchange student in Electrical Engineering Department

RESEARCH INTERESTS

In short: I work on system-level optimization for AI/LLM systems, aiming to make them cheaper, faster, more reliable, and more capable.

My current research focuses on resource optimization for LLM agent systems under API budget and latency constraints, as well as efficient distributed learning systems. Topics include parameter-efficient fine-tuning for LLMs [2]; online learning for hierarchical inference [5]; efficient LLM deployment [1], [4]; federated learning [3], [6]–[10]; and medical AI [11].

TEACHING AND MENTORING

Teaching Assistant, 18-786 Introduction to Deep Learning (CMU Spring 2025), 05695 Math Foundations of AI (UT-Austin Fall 2026)

CMU Peer Mentor (Spring 2025), mentored Mary Tonwe, Fanyun Xu, and Han Zhou

ACHIEVEMENTS

IEEE-Eta Kappa Nu (HKN) member, Best Poster Award ICDCS 2024, Outstanding Graduate (HUST)

SKILLS

Pytorch, Tensorflow, Sklearn, Pandas, Numpy, SGLang, verl, LangChain.

PUBLICATIONS

- [1] **H. Zhang**, D. Kim, S. Cha, K. S. Chan, A. Swami, G. D. Veciana, and H. Vikalo, “Jove: Joint execution and verification for resource-aware llm task graphs,” *arXiv preprint arXiv:2610.03296*,
- [2] **H. Zhang**, D. Kim, S. Cha, and H. Vikalo, “FedRot-LoRA: Mitigating rotational misalignment in federated lora,” in *International Conference on Machine Learning (ICML)*, 2026.
- [3] **H. Zhang**, C. F. Pereira, M. Siew, X. Liu, C. Joe-Wong, and R. El-Azouzi, “FedSteer: Taming extreme gradient staleness in federated learning with corrective projections and caching,” in *42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.
- [4] S. Cha, H. Chen, D. Kim, **H. Zhang**, K. Chan, G. de Veciana, and H. Vikalo, “CoreQ: Learning-free mismatch correction and successive rounding for quantization,” in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, **spotlight**, 2026.

- [5] **H. Zhang**, S. Cha, H. B. Beytur, K. S. Chan, G. de Veciana, and H. Vikalo, “Online learning for multi-layer hierarchical inference under partial and policy-dependent feedback,” *arXiv preprint arXiv:2603.04247*, 2026.
- [6] **H. Zhang**, “Optimally leveraging stale updates in federated learning,” *ACM SIGMETRICS Performance Evaluation Review*, vol. 53, no. 3, pp. 20–22, 2026.
- [7] M. Siew, **H. Zhang**, J.-I. Park, Y. Liu, Y. Ruan, L. Su, S. Ioannidis, E. Yeh, and C. Joe-Wong, “Fair concurrent training of multiple models in federated learning,” *IEEE Transactions on Networking*, 2025.
- [8] Z. Gong, **H. Zhang**, M. Siew, C. Joe-Wong, and R. El-Azouzi, “Group-based client sampling in multi-model federated learning,” in *2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring)*, IEEE, 2025, pp. 1–7.
- [9] **H. Zhang**, Z. Gong, Z. Li, M. Siew, C. Joe-Wong, and R. El-Azouzi, “Towards optimal heterogeneous client sampling in multi-model federated learning,” *arXiv preprint arXiv:2504.05138*, 2025.
- [10] **H. Zhang**, Z. Li, Z. Gong, M. Siew, C. Joe-Wong, and R. El-Azouzi, “Poster: Optimal variance-reduced client sampling for multiple models federated learning,” in *2024 IEEE 44th International Conference on Distributed Computing Systems (ICDCS)*, **Best Poster Award**, IEEE, 2024.
- [11] **H. Zhang** and H. Chen, “Efficient 3d transformer with cluster-based domain-adversarial learning for 3d medical image segmentation,” in *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, IEEE, 2023, pp. 1–5.